Linwei YU

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EDUCATION

University of California at Berkeley, Berkeley School of Education

Degree: *M.A. in Education, Social Research Methodologies cluster* CA, starting in 08/2024

Advisors: *PROF. Michelle Wilkerson* and *PROF. Mark Wilson*

Honors and Awards: MA Scholarship for the 2024-25 Academic Year (one-time award of $5,000)

University of Macau, Faculty of Education

Degree: *B.Ed. in Integrated Science, with specialization in Chemistry* Macau, 09/2020 – 06/2024

Grade: 3.57/4.00 (Ranking in major: 1st out of 10)

Honors: Dean's Honour List 2020–2021, 2021–2022, 2022–2023

Core Courses: Emerging Pedagogies of STEM Teaching (A), Assessment in Science Education (A), Science Education in Secondary Schools (A), Scientific Inquiry and Practical Work in Secondary Schools (A), Educational Research (A), etc.

Fudan University

Hong Kong and Macao Ten Thousand Program_Autumn Course Online, 09/2022 – 12/2022

Course Taken: Probability and Statistics

University of Cambridge, Institute of Continuing Education

Virtual Summer Festival of Learning course Online, 07/2021

Course Taken: The Psychology of Adolescence: Brain and Behaviour

RESEARCH INTERESTS

- STEM curriculum, pedagogies, and assessment.
- Computational thinking (CT), and technology-enhanced learning.

PUBLICATIONS

Wei, B., Zhan, Z., Jiang, Z., & **Yu, L.** (2024). Representation of Learning Outcomes Stipulated by the Intended Curriculum in Four Series of Chemistry Textbooks: Based on Legitimation Code Theory. *Research in Science Education*. (Accepted for revision, submission ID: RISE-D-24-00019) (SSCI)

Qiu, R., Wang, B., Shang, J., Hu, G., **Yu, L.,** & Gao, X. (2024). Modifying hollow glass microspheres to obtain self-floating separation adsorbents for adsorbing pollutants in wastewater: A review. *Journal of Molecular Liquids, 404,* 124965. https://doi.org/10.1016/j.molliq.2024.124965. (SCI)

(Paper in Progress)

Yu, L., Li, Z., & Oon, P.T. (XX). A Pedagogical Framework to Foster Computational Thinking in STEM: From the Perspective of Teachers.

Yu, L. & Wei, B. (XX). The Meaning of 'Experiment' in the Chemistry Curriculum Standards in China: A Content Analysis.

ACADEMIC RESEARCH EXPERIENCE

A Pedagogical Framework to Foster Computational Thinking in STEM

University of Macau, Educational Research Center — Macau, 03/2023 – Present

Research Project | Advisor: PROF. Emily Pey-Tee OON

- Collaborating with a Ph.D. student to conduct research related to computational thinking (CT), including curriculum design, pedagogical framework design aimed at improving students' computational thinking ability. Writing a paper titled "A Pedagogical Framework to Foster Computational Thinking in STEM: From the Perspective of Teachers" as the first author.

The Meaning of "Experiment" in Junior and Senior High School Science Curriculum Standards in China

University of Macau, Faculty of Education — Macau, 11/2022 – Present

Research Project | Advisor: PROF. Bing WEI

- Conducting research on the meaning of "experiment" in junior and senior high school science curriculum standards in China. Conducting content analysis, managing and analyzing data, and writing a paper titled "The Meaning of 'Experiment' in the Chemistry Curriculum Standards in China: A Content Analysis" as the first author.

A Study of Primary Teachers' Perceptions about Science Curriculum in China;
The Representation of Learning Outcomes Stipulated by the Intended Curriculum in Chemistry Textbooks

University of Macau, Faculty of Education — Macau, 05/2023 – 07/2023

Summer Research Programme | Principal Investigator: PROF. Bing WEI

- Engaged in summer research programme, analyzed the representation of learning outcomes stipulated by the intended curriculum in four series of chemistry textbooks, based on legitimation code theory. Conducted data analysis and drafted the results part of paper (co-author).

Preparation of Self-suspension Adsorption Functional Materials by Modification of Hollow Glass Microspheres

Southwest Petroleum University, School of Chemistry and Chemical Engineering — Sichuan, 06/2022 – 07/2022

Assisted research | Principal Investigator: PROF. Bing WANG

- Assisted research on the production of self-suspension adsorption functional materials by modifying hollow glass microspheres. Conducted experiments, managed data, drafted and proofread the review paper (co-author).

A Comparative Study of Technological and Engineering Education Policies in China and the USA

University of Macau, Faculty of Education — Macau, 05/2022 – 06/2022

Assisted research | Principal Investigator: PROF. Bing WEI

- Assisted STEM education policy research of the *Standards for Technological and Engineering Literacy* published in the United States (ITEEA, 2020), and *The Science Curriculum Standards of Compulsory Education (the 2020 version)* in China (MoE, 2020). Conducted literature review and conducted content analysis.

TEACHING EXPERIENCE

Supervised Teaching and School Experience, Chemistry and Scientific Inquiry Teacher

Macau Baptist College (Escola Cham Son de Macau) — Macau, 10/2023 – 12/2023

Supervisor: PROF. Bing WEI | Mentor: MS. Ka-Ming TSANG

- Taught junior high school grade 2 and grade 3 Chemistry, senior high school grade 2 Scientific Inquiry, specific works including giving lectures, teaching lab classes, assessment, *etc.*

SERVICE

Student Representative, Programme Management Committee of Integrated Science Programme (PMC-INS), University of Macau Macau, 06/2022 – Present

12th PR Student Ambassador, University of Macau Macau, 03/2022 – Present

House Leader, House of Integrated Science, University of Macau Macau, 08/2022 – 08/2023

Award: Academically Outstanding House Award 2022/2023; The Most Active House Award 2022/2023

Secretary, Drama Society, University of Macau Macau, 10/2021 – 10/2023

Associate House Leader, House of Integrated Science, University of Macau Macau, 08/2021 – 08/2022

ADDITIONAL INFORMATION

Software and Tools: SPSS, Winsteps, ENA Web Tool, Scratch 3.0, mBlock, MATLAB, Arduino

Language Proficiency: Chinese (Mandarin: Native; Cantonese: Listening and Writing), English (IELTS: Overall 7.0)